The file’s structure dominates its label associations

- 22,457 rows, 17 columns; source is undocumented.
- After removing the ordered block, smoking OR is 1.00.
- Interpretation: integrity audit and descriptive analysis only.

A smoking checkbox cannot reproduce screening history

- USPSTF 2021 [1] uses ages 50–80.
- It also requires at least 20 pack-years and a quit-time criterion of 15 years.
- This file has no dose, duration, or quit-time fields; no eligibility inference is possib

The objective is to test whether findings survive integrity checks

- RQ1: characterize ordering, duplicates, and conflicting records.
- RQ2–RQ3: quantify associations and their sensitivity to cleaning.
- RQ4–RQ5: describe representativeness limits and justify candidate features.
- Success means reproducible checks, transparent uncertainty, and traceable claim

The checksum does not match the project contract

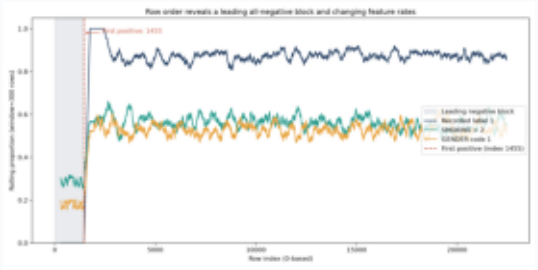
- Observed: 22,457 × 17; 0 null cells.
- Expected SHA-256 prefix: 4ad2333067fcb0a9.
- Observed prefix: 1e0d557990cd7533.
- Recorded-label code 1 share: 82.1%; not population prevalence.

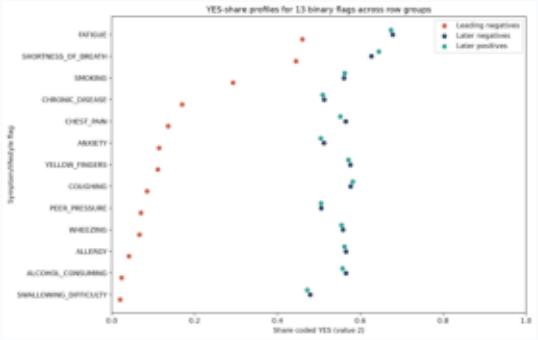
Several meanings needed for interpretation are unknown

- Dataset source, sampling frame, collection process, and licence: unknown.
- Meaning and generation of the recorded label: unknown.
- GENDER code mapping: unknown.
- The project README only supplied a title; redistribution needs a licence check.

The leading rows form an all-negative block

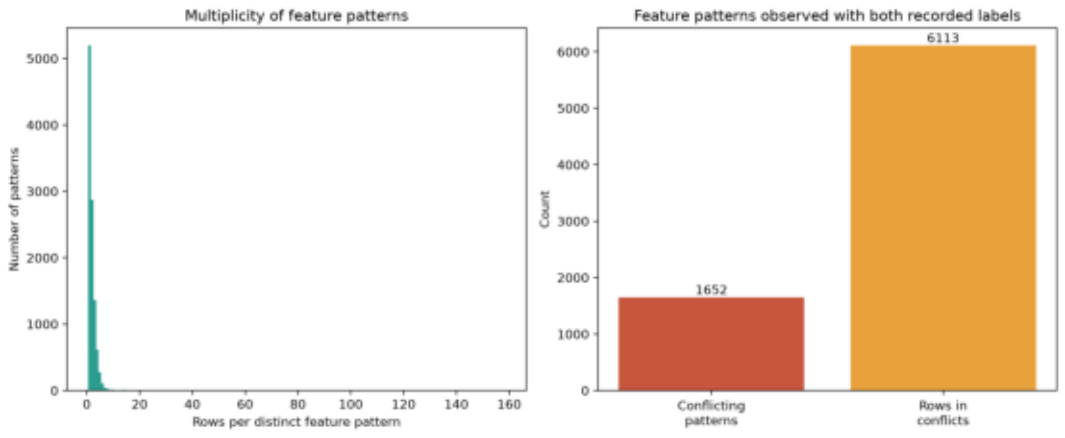
- First 1,455 rows are coded negative.
- Smoking YES share: 29.3% in the block vs 56.0% in later negatives.
- The plot shows row order, not a documented timeline.





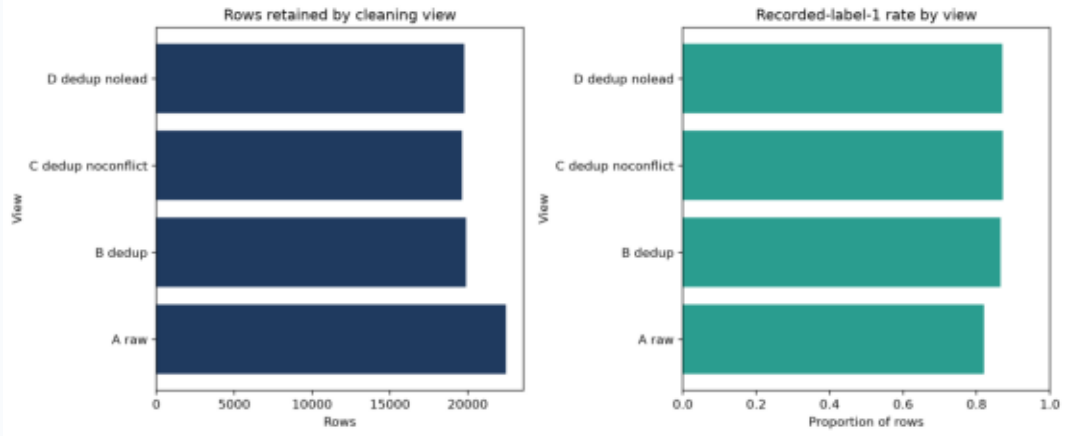
Duplicate patterns and ID order are not harmless

- 2,573 exact duplicate rows; 133 conflicting feature patterns.
- Maximum pattern multiplicity: 86.
- ID-only AUC: 0.67 — an audit statistic, not model performance.
- The source and record-generation process remain unknown.



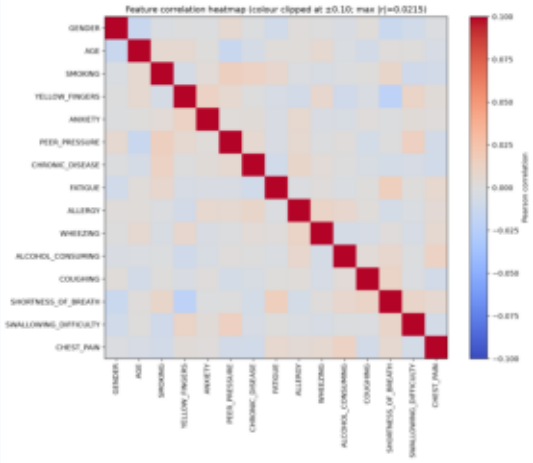
Four cleaning views expose policy sensitivity

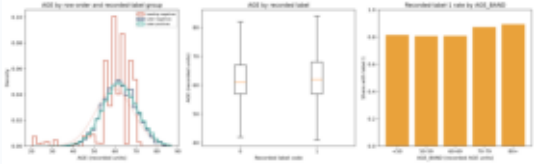
- A raw: 22,457; B exact-deduplicated: 19,884.
- C no conflicts: 19,618; D no leading block: 19,781.
- Recorded-positive share ranges from 82.1% to 87.1%.
- View D filters B by the original raw row index.



After deduplication, pairwise feature relations are weak

- Age: 21–87, mean 62.1.
- Maximum absolute pairwise correlation: 0.022.
- First PCA component: 7.0%; flat reference: 6.7%.
- These are descriptive diagnostics, not proof of independence.



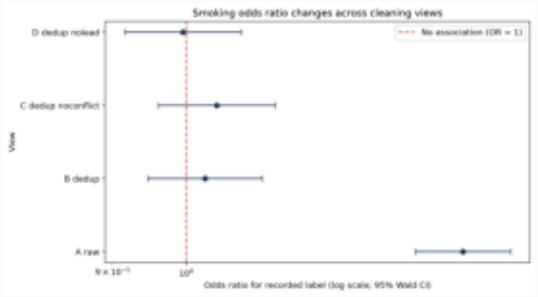


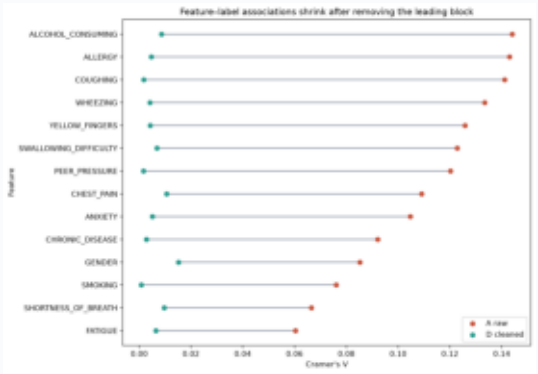
Five pre-specified tests separate effect from significance

- H1 smoking: odds ratio, confidence interval, chi-square, permutation check.
- H2 age: Mann–Whitney and CLES; H3 symptom count: Spearman rho.
- H4 feature family: Benjamini–Hochberg correction; H5: source-shift audit.
- View D H4 minimum raw p = 0.033; minimum q = 0.458.

The smoking association moves toward the null after cleaning

- View A OR 1.49 (1.39–1.59).
- View D OR 1.00 (0.92–1.08).
- View D H4 minimum q = 0.46.
- The file does not establish a population or causal conclusion.





Feature construction is transparent, not evidence of prediction

- F0: 15 raw features; F1: 19 with selected descriptors.
- F2: 5 domain-prior candidates.
- Age scaling must be fitted inside training folds in any future model.
- The smoking-by-age feature is a proxy, not screening eligibility.

The file supports an integrity finding, not a clinical claim

- After removing the leading block, smoking OR is 1.00.
- After BH correction in D, the minimum q is 0.46.
- Provenance, label meaning, representativeness, and external validity are unresolv
- All conclusions are limited to this supplied file.

Next steps are planned, not completed

- Verify source, licence, checksum, label definition, and sampling frame.
- If authorized, plan grouped/source-aware validation on independent data.
- Select metrics and calibration checks with the intended use in mind.
- Clean-clone testing and timed presentation rehearsal remain to be done.